Introductory Calculus

ANSWER KEY

Section 1: Limits & Continuity

1 Evaluate $\lim_{x\to 2} \frac{x^3-8}{x-2}$.

a. 12 ✓

b. 8

c. 4

d. Does not exist

Factor difference of cubes: $\lim_{x\to 2} \frac{(x-2)(x^2+2x+4)}{x-2} = \lim_{x\to 2} (x^2+2x+4) = 4+4+4 = 12$.

2 Evaluate $\lim_{x\to 0} \frac{\tan(3x)}{x}$.

a. 3 ✓

b. 1

c. $\frac{1}{3}$

d. 0

$\lim_{x\to 0} \frac{\sin(3x)}{x\cos(3x)} = 3\lim_{x\to 0} \frac{\sin(3x)}{3x} \cdot \frac{1}{\cos(3x)} = 3(1)(1) = 3$.

3 What value of a makes $f(x) = \frac{x^2-1}{x-1}$ for $x \neq 1$, and $f(1) = a$ continuous at $x = 1$?

a. 2 ✓

b. 1

c. 0

d. -1

$\lim_{x\to 1} \frac{(x-1)(x+1)}{x-1} = \lim_{x\to 1} (x+1) = 2$. For continuity, $f(1) = a = 2$.

Section 2: Derivatives & Power/Product Rules

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4 Find the derivative of $f(x) = 4x^3 - 5x^2 + 7x - 9$.

a. $12x^2 - 10x + 7$ ✓

b. $12x^2 - 5x + 7$

c. $4x^2 - 10x + 7$

d. $12x^3 - 10x^2 + 7$

By power rule: $\frac{d}{dx}(4x^3) = 12x^2$, $\frac{d}{dx}(-5x^2) = -10x$, $\frac{d}{dx}(7x) = 7$.

5 What is the slope of the tangent line to $y = x^2e^x$ at $x = 1$?

a. $3e$ ✓

b. $2e$

c. e

d. $4e$

Product rule: $y' = 2xe^x + x^2e^x = e^x(2x + x^2)$. At $x = 1$: $y'(1) = e^1(2 + 1) = 3e$.

6 What is the derivative of $g(x) = \frac{2x+1}{x-3}$?

a. $-\frac{7}{(x-3)^2}$ ✓

b. $\frac{7}{(x-3)^2}$

c. $\frac{2}{(x-3)^2}$

d. $-\frac{5}{(x-3)^2}$

Quotient rule: $\frac{2(x-3)-(2x+1)(1)}{(x-3)^2} = \frac{2x-6-2x-1}{(x-3)^2} = -\frac{7}{(x-3)^2}$.

Section 3: Chain Rule & Extrema

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7 Find the derivative of $h(x) = (3x^2 - 5)^4$.

a. $24x(3x^2 - 5)^3$ ✓

b. $4(3x^2 - 5)^3$

c. $12x(3x^2 - 5)^3$

d. $24(3x^2 - 5)^3$

Chain rule: $4(3x^2 - 5)^3 \cdot \frac{d}{dx}(3x^2 - 5) = 4(3x^2 - 5)^3(6x) = 24x(3x^2 - 5)^3$.

8 At what value of x does $f(x) = 2x^3 - 9x^2 + 12x + 1$ have a local minimum?

a. $x = 2$ ✓

b. $x = 1$

c. $x = 0$

d. $x = 3$

$f'(x) = 6x^2 - 18x + 12 = 6(x^2 - 3x + 2) = 6(x - 1)(x - 2) = 0$. $f''(x) = 12x - 18$. At $x = 2$: $f''(2) = 6 > 0$ (local min).

9 What is the acceleration of a particle at $t = 2$ if its position is $s(t) = t^3 - 6t^2 + 5t$?

a. 0 ✓

b. 5

c. -7

d. 6

Velocity $v(t) = s'(t) = 3t^2 - 12t + 5$. Acceleration $a(t) = v'(t) = 6t - 12$. At $t = 2$: $a(2) = 6(2) - 12 = 0$.

Section 4: Definite Integrals & Fundamental Theorem

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10 Evaluate the definite integral: $\int_0^3 (3x^2 - 2x + 1)dx$.

a. 21 ✓

b. 18

c. 24

d. 27

Antiderivative is $[x^3 - x^2 + x]_0^3 = (27 - 9 + 3) - 0 = 21$.

11 What is the area of the region bounded by $y = 4 - x^2$ and the x-axis?

a. $\frac{32}{3}$ ✓

b. $\frac{16}{3}$

c. 16

d. 8

Roots are $x = \pm 2$: $\int_{-2}^{2}(4 - x^2)dx = 2\int_0^2 (4 - x^2)dx = 2[4x - \frac{x^3}{3}]_0^2 = 2[8 - \frac{8}{3}] = 2[\frac{16}{3}] = \frac{32}{3}$.

12 If $\int_1^4 f(x)dx = 7$, what is $\int_1^4 [2f(x) + 3]dx$?

a. 23 ✓

b. 17

c. 20

d. 26

$2\int_1^4 f(x)dx + \int_1^4 3dx = 2(7) + 3(4 - 1) = 14 + 9 = 23$.

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